

CSE5DMI Data Mining

Week 07
ASSOCIATION ANALYSIS: BASIC CONCEPTS AND ALGORITHMS

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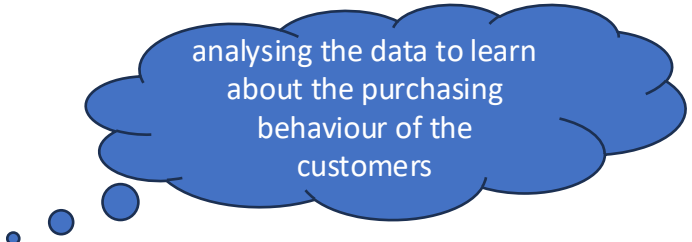
WEEK 7 TOPICS

- Association rule mining basics
- Frequent itemset generation
- Rule generation (Week8)

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Association Rule Mining

Given a set of transactions, find rules that will predict the occurrence of an item based on the occurrences of other items in the transaction



Market-Basket transactions

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Association Rules

{Diaper} → {Beer},
{Milk, Bread} → {Eggs,Coke},
{Beer, Bread} → {Milk},

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Definition: Frequent Itemset

Itemset

- A collection of one or more items
  - Example: {Milk, Bread, Diaper}
- k-itemset
  - An itemset that contains k items

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

e.g. 2-itermset

The last transaction contains an itemset {break, Coke}, but it does not contain an itemset of {bread, eggs}

Support count (σ)

- the number of transactions that contain a particular itemset
- E.g. σ({Milk, Bread,Diaper}) = 2 because in the table, there are two transactions contain such an itemset

Support

- Fraction of transactions that contain an itemset = Support count / number of transactions
- E.g. s({Milk, Bread, Diaper}) = 2/5

Frequent Itemset

- An itemset whose support is greater than or equal to a *minsup* threshold

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Definition: Association Rule

- Association Rule
  - An implication expression of the form $X \rightarrow Y$, where X and Y are disjoint itemsets
  - Example:
    {Milk, Diaper} → {Beer}

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

- Rule Evaluation Metrics
  - Support (s)
    - Fraction of transactions that contain both X and Y
  - Confidence (c)
    - Measures how often items in Y appear in transactions that contain X

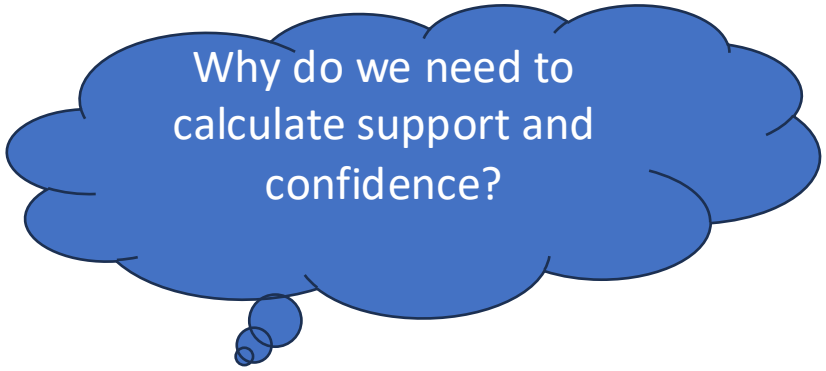
Example:

$$\{\text{Milk}, \text{Diaper}\} \Rightarrow \{\text{Beer}\}$$

$$s = \frac{\sigma(\text{Milk}, \text{Diaper}, \text{Beer})}{|T|} = \frac{2}{5} = 0.4$$

$$c = \frac{\sigma(\text{Milk}, \text{Diaper}, \text{Beer})}{\sigma(\text{Milk}, \text{Diaper})} = \frac{2}{3} = 0.67$$

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Support is an important measure because a rule that has very low support might occur simply by chance.

From business perspective, a low support rule is unlikely to be interesting because it might not make profit by promoting items that customers seldom buy together.

we are interested in finding rules whose support is greater than a user-defined threshold.

Confidence measures the reliability of the inference made by a rule. The higher the confidence, the more likely it is for Y to be present in transactions that contain X

Confidence also provides an estimate of the conditional probability of Y given X.

Note: The inference made by an association rule *does not necessarily imply relationship between cause and effect*. Instead, it can sometimes suggest a strong co-occurrence relationship between items in the X and consequent of the rule.

How to formulate Association Rule Mining problem

Given a set of transactions T, the goal of association rule mining is to find all rules having

- support ≥ *minsup* threshold
- confidence ≥ *minconf* threshold

One approach is Brute-force:

- List all possible association rules
- Compute the support and confidence for each rule
- Prune rules that fail the *minsup* and *minconf* thresholds

⇒ Majority of them are discarded at a later stage: Computationally prohibitive!

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Mining Association Rules

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

A simple way to improve the performance of association rule mining algorithm is to decouple the support and confidence requirements.

- Observation:

Support of Rule X->Y = Support of itemset X U Y

e.g. Rules

{Milk,Diaper} → {Beer} (s=0.4, c=0.67)
{Milk,Beer} → {Diaper} (s=0.4, c=1.0)
{Diaper,Beer} → {Milk} (s=0.4, c=0.67)
{Beer} → {Milk,Diaper} (s=0.4, c=0.67)
{Diaper} → {Milk,Beer} (s=0.4, c=0.5)
{Milk} → {Diaper,Beer} (s=0.4, c=0.5)

Rule Evaluation Metrics

3

- Support (s)
  - Fraction of transactions that contain both X and Y
- Confidence (c)
  - Measures how often items in Y appear in transactions that contain X

If the itemset is infrequent, then all six candidate rules can be pruned and discarded immediately.

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Mining Association Rules

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Observations:

- All the above rules are binary partitions of the same itemset: {Milk, Diaper, Beer}
- Rules originating from the same itemset have identical support but can have different confidence
- Thus, a common strategy adopted by many association rule mining algorithms is to decompose the problem into two major subtasks.

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Mining Association Rules

Two-step approach:



1. Frequent Itemset Generation
   - Generate all itemsets whose support ≥ minsup (user-defined support threshold)



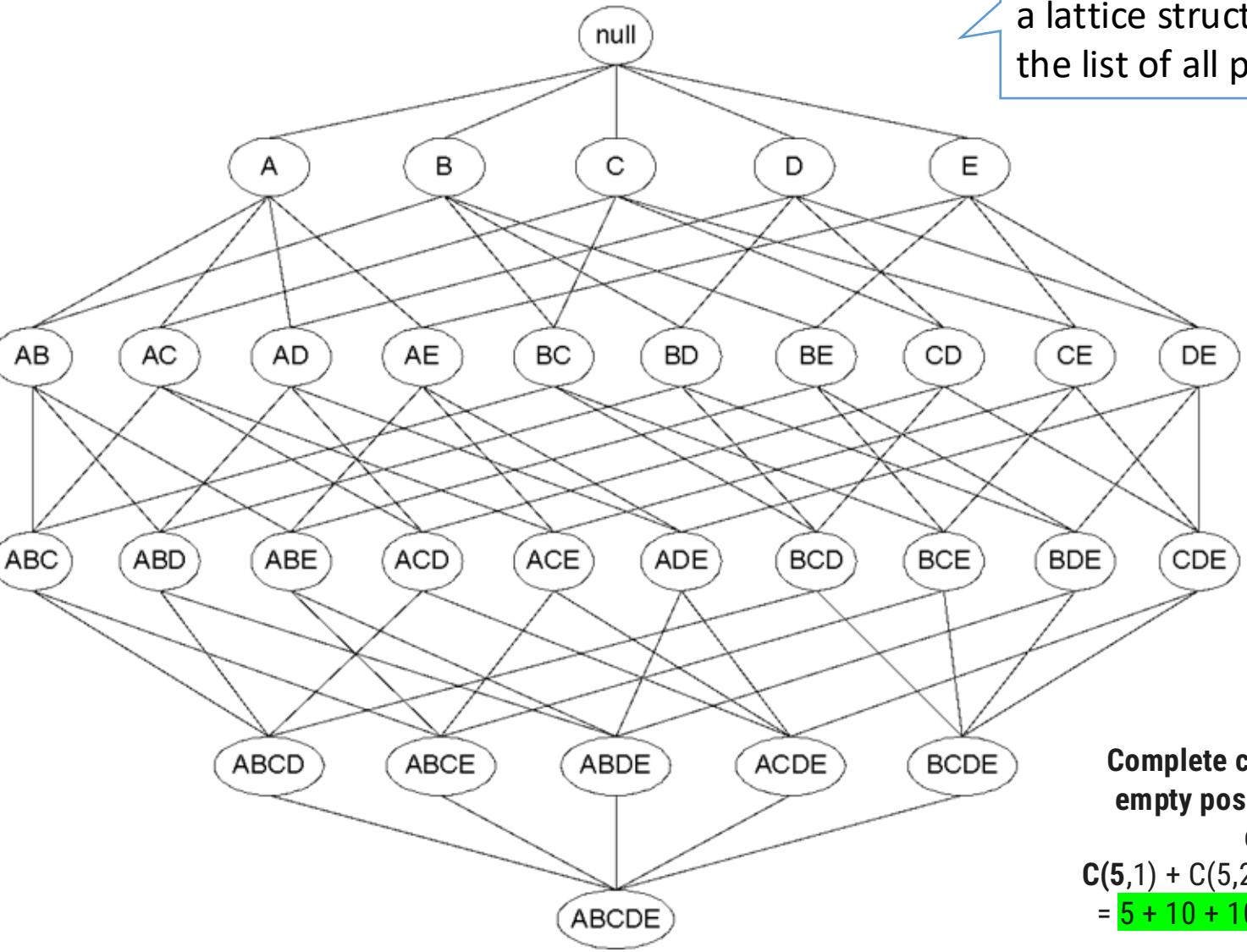
2. Rule Generation
   - Generate high confidence rules from each frequent itemset, where each rule is a binary partitioning of a frequent itemset
   - The aim of this step is to extract all the rules with high confidence from the frequent itemsets obtained in the first sub-step

Frequent itemset generation is still computationally expensive

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Frequent Itemset Generation

a lattice structure to enumerate the list of all possible itemsets



For five items A, B, C, D and E.

Level 1 contains 1-itemsets, level 2 contains pairs, level 3 contains triplets, and so on until ABCDE.

Complete candidate space (all non-empty possible itemsets). For d=5 distinct items

C(5,1) + C(5,2) + C(5,3) + C(5,4) + C(5,5)

= 5 + 10 + 10 + 5 + 1 = 31 = 2^5 - 1

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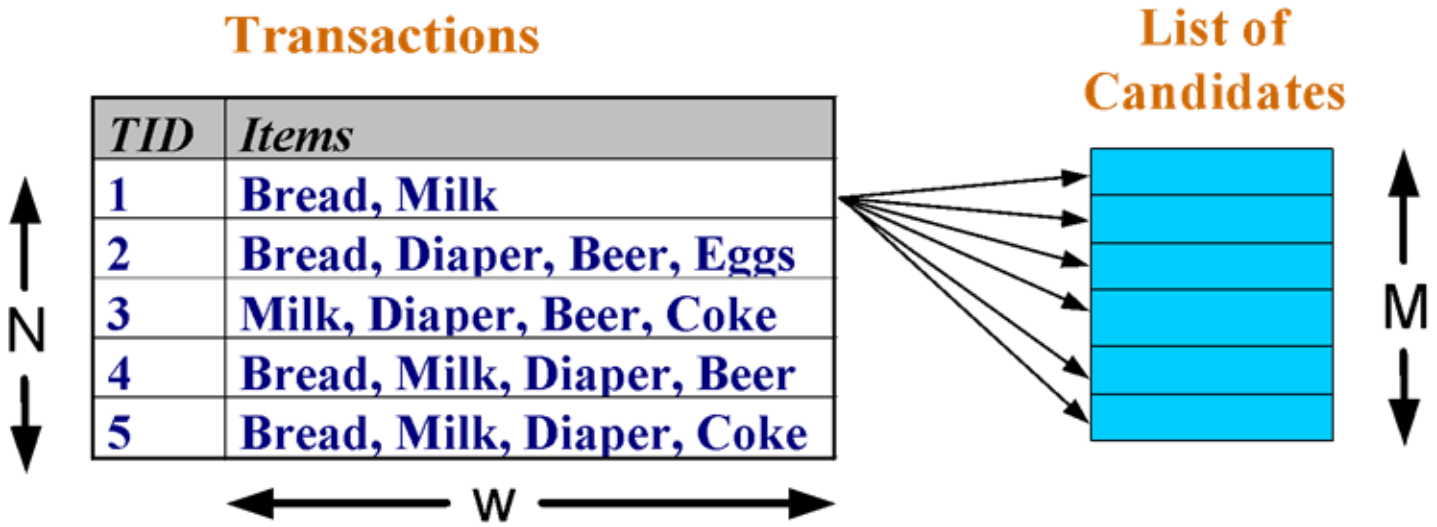
Frequent Itemset Generation

Brute-force approach:

N transactions and
M candidate itemsets
d = 6 (unique)
w=4: maximum number of items in a single transaction.

```
for each transaction N:
    for each candidate itemset M:
        if every item in M appears in N:
            support_count[C] += 1
```

- Each itemset in the lattice is a candidate frequent itemset
- Count the support of each candidate by scanning the database



- Match each transaction against every candidate
- Complexity ~ O(NMw) => Expensive since M = 2^d !!!

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Why Transaction Width w Matters

Candidate does not match

T4 = {Bread, Milk, Diaper, Beer}

C = {Milk, Beer, Eggs}

Milk is present: yes

Beer is present: yes

Eggs is present: no

Because Eggs is absent, $C \not\subset T4$.

The support count does not increase.

Candidate matches

T4 = {Bread, Milk, Diaper, Beer}

C = {Milk, Beer, Diaper}

Milk is present: yes

Beer is present: yes

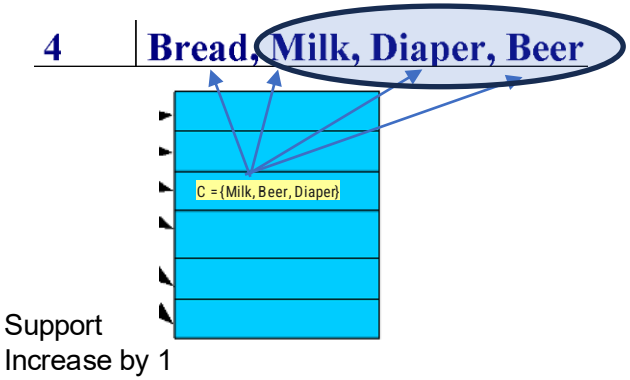
Diaper is present: yes

All items are present, so $C \subseteq T4$.

The support count increases by 1.

Why w matters

A comparison may inspect up to w items. Across N transactions and M candidates: O(NMw).



If all items in the itemset's candidate is a subset of T4

the algorithm must check whether every candidate item is present in the transaction

T4 is compared to one of candidate itemset

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Worked Example: Transactions and Candidate Itemsets

Transaction	Products purchased
T1	Bread, Milk
T2	Bread, Diaper, Beer, Eggs
T3	Milk, Diaper, Beer
T4	Bread, Milk, Diaper, Beer
T5	Bread, Milk, Diaper, Coke

Candidate	Itemset
C1	{Bread, Milk}
C2	{Diaper, Beer}
C3	{Milk, Beer}
C4	{Bread, Eggs}

Complete candidate space (all non-empty itemsets)

C(6,1) + C(6,2) + C(6,3) + C(6,4) + C(6,5) + C(6,6)

= 6 + 15 + 20 + 15 + 6 + 1 = 63 = 2^6 - 1

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Worked Example: Match Transactions to Candidates

Transaction	Bread, Milk	Diaper, Beer	Milk, Beer	Bread, Eggs
T1	✓	✗	✗	✗
T2	✗	✓	✗	✓
T3	✗	✓	✓	✗
T4	✓	✓	✓	✗
T5	✓	✗	✗	✗
Support count	**3**	**3**	**2**	**1**

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Worked Example: T4 Check and Support

T4 = {Bread, Milk, Diaper, Beer}

{Bread, Milk} matches: both products are present.

{Diaper, Beer} matches.

{Milk, Beer} matches.

{Bread, Eggs} does not match: Eggs are absent.

s({Bread, Milk}) = 3/5 = 60%

s({Diaper, Beer}) = 3/5 = 60%

s({Milk, Beer}) = 2/5 = 40%

s({Bread, Eggs}) = 1/5 = 20%

Minimum support = 50%

Frequent itemsets: {Bread, Milk} and {Diaper, Beer}

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Frequent Itemset Generation Strategies

Reduce the number of candidates (M)

- Complete search: $M=2^d$
- Use pruning techniques to reduce M

Reduce the number of comparisons (NM)

- Use efficient data structures to store the candidates or transactions
- No need to match every candidate against every transaction

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Reducing Number of Candidates

- Apriori principle:
  - Definition: If an itemset is frequent, then all of its subsets must also be frequent
    - Or, conversely, if an itemset is infrequent, then all its supersets must be infrequent too
  - If we found the itemset {a,b} is infrequent, then the entire subgraph containing the supersets of {a,b} can be pruned immediately.

- Apriori principle holds due to the following property of the support measure:

$$\forall X, Y : (X \subseteq Y) \Rightarrow s(X) \geq s(Y)$$

  - Support of an itemset never exceeds the support of its subsets
  - This is known as the anti-monotone property of support

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More detailed explanation

The formula

$$\forall X,Y: (X \subseteq Y) \Rightarrow s(X) \geq s(Y)$$

∀ means “for every.”

X and Y are itemsets.

$X \subseteq Y$ means every item in X is also in Y.

⇒ means “implies.”

Support notation

s(X) the support of itemset X.

s(Y) the support of itemset Y.

≥ “greater than or equal to.”

Plain-language meaning

If X is a subset of Y, then X must occur at least as often as Y.

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Why Support Cannot Increase

Example itemsets

X = {a, b}

Y = {a, b, c}

Every transaction containing Y automatically contains X.

However, a transaction may contain {a, b} without containing c.

Transaction-set relationship

Transactions(Y) ⊆ Transactions(X)

Therefore

s({a,b}) ≥ s({a,b,c})

Example: 60% ≥ 40%

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Apriori Pruning Rules - Example

If Y is frequent, every **subset of Y** must also be frequent.

Example

If {a,b,c} is frequent, then these must be frequent:

{a}, {b}, {c}

{a,b}, {a,c}, {b,c}

Each subset has support **at least as large as s(Y).**

Pruning example

Minimum support = 50%

s({a,b}) = 40%

Because {a,b} is infrequent, every candidate containing both a and b can be removed immediately.

Their supports do not need to be calculated.

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Illustrating Apriori Principle

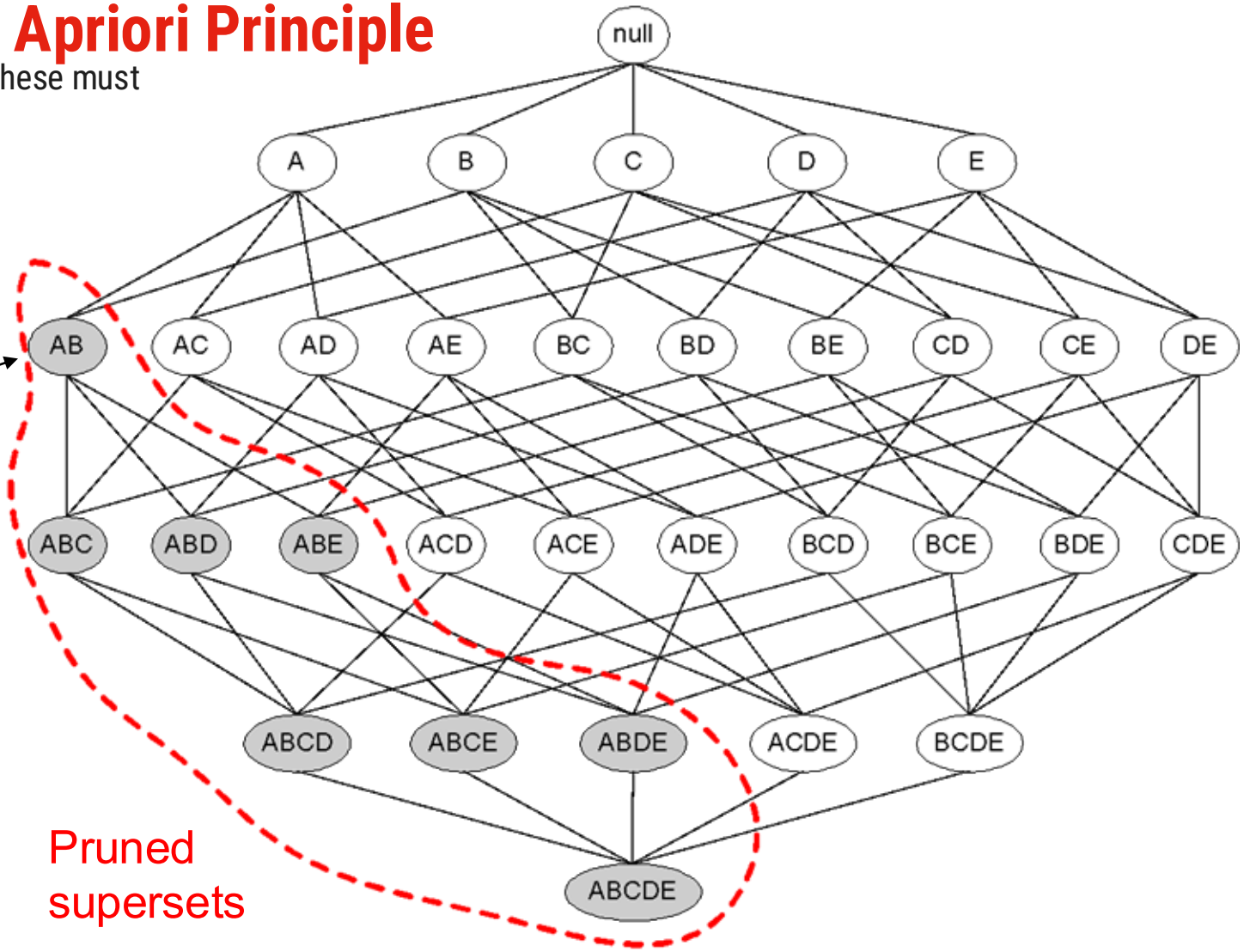
If {a,b} is frequent, then these must be frequent:

{a,b,c}, {a,b,d}, {a,b,e}

{a,b,c},{a,b,c,e},{a,b,d,e}

{a,b,c,d,e}

Found to be Infrequent



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Illustrating Apriori Principle

TID	Items
1	Bread, Milk
2	Beer, Bread, Diaper, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Bread, Coke, Diaper, Milk



Items (1-itemsets)

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Minimum Support = 3

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Illustrating Apriori Principle

TID	Items
1	Bread, Milk
2	Beer, Bread, Diaper, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Bread, Coke, Diaper, Milk



Items (1-itemsets)

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Minimum Support = 3

We discard the itemset whose support count is smaller than 3, and only keep frequent itemsets

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Illustrating Apriori Principle

TID	Items
1	Bread, Milk
2	Beer, Bread, Diaper, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Bread, Coke, Diaper, Milk

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Items (1-itemsets)

Itemset
{Bread,Milk}
{Bread, Beer }
{Bread,Diaper}
{Beer, Milk}
{Diaper, Milk}
{Beer,Diaper}

Pairs (2-itemsets)

(No need to generate candidates involving Coke or Eggs)

Minimum Support = 3

Continue to 2-itemset.

- Only use the frequent 1-itemsets because of the apriori principle
- Discard the itemset whose support count is smaller than 3, and only keep frequent itemsets

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Illustrating Apriori P

TID	Items
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2	Beer, Bread, Diaper, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Bread, Coke, Diaper, Milk

Items (1-itemsets)

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Itemset	Count
{Bread,Milk}	3
{Beer, Bread}	2
{Bread,Diaper}	3
{Beer,Milk}	2
{Diaper,Milk}	3
{Beer,Diaper}	3

Pairs (2-itemsets)

(No need to generate candidates involving Coke or Eggs)

Minimum Support = 3

We discard the itemset whose support count is smaller than 3, and only keep frequent itemsets

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Illustrating Apriori P

TID	Items
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Items (1-itemsets)

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Itemset	Count
{Bread,Milk}	3
{Bread,Beer}	2
{Bread,Diaper}	3
{Milk,Beer}	2
{Milk,Diaper}	3
{Beer,Diaper}	3

Pairs (2-itemsets)

(No need to generate candidates involving {Bread,Beer} or {Milk,Beer})

Minimum Support = 3

Triplets (3-itemsets)

Itemset
{ Beer, Diaper, Milk}
{ Beer,Bread,Diaper}
{Bread,Diaper,Milk}
{ Beer, Bread, Milk}

With the apriori principle, we only need to keep candidate 3-itemsets whose subsets are frequent. Only 1

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Illustrating Apriori P

TID	Items
1	Bread, Milk
2	Beer, Bread, Diaper, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Bread, Coke, Diaper, Milk

Items (1-itemsets)

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Itemset	Count
{Bread,Milk}	3
{Bread,Beer}	2
{Bread,Diaper}	3
{Milk,Beer}	2
{Milk,Diaper}	3
{Beer,Diaper}	3

Pairs (2-itemsets)

(No need to generate candidates involving {Bread,Beer} or {Milk,Beer})

Minimum Support = 3

Triplets (3-itemsets)

Itemset	Count
{ Beer, Diaper, Milk}	2
{ Beer,Bread, Diaper}	2
{Bread, Diaper, Milk}	2
{Beer, Bread, Milk}	1

No need to test 4-itemset

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Illustrating Apriori Principl

TID	Items
1	Bread, Milk
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3	Beer, Coke, Diaper, Milk
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Items (1-itemsets)

Item	Count
Bread	4
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Beer	3
Diaper	4
Eggs	1

Itemset	Count
{Bread,Milk}	3
{Bread,Beer}	2
{Bread,Diaper}	3
{Milk,Beer}	2
{Milk,Diaper}	3
{Beer,Diaper}	3

Pairs (2-itemsets)

(No need to generate candidates involving {Bread,Beer} or {Milk,Beer})

Minimum Support = 3

Triplets (3-itemsets)

Itemset	Count
{ Beer, Diaper, Milk}	2
{ Beer,Bread, Diaper}	2
{Bread, Diaper, Milk}	2
{Beer, Bread, Milk}	1

If every subset is considered,

$${}^6C_1 + {}^6C_2 + {}^6C_3$$

$$6 + 15 + 20 = 41$$

With support-based pruning,

$$6 + 6 + 1 = 13$$

Searching complexity is reduced by about 70%

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Apriori Algorithm: Summary

- F_k: frequent k-itemsets
- L_k: candidate k-itemsets

- Algorithm
  - Let k=1
  - Generate F_1 = {frequent 1-itemsets}
  - Repeat until F_k is empty
    - **Candidate Generation**: Generate L_{k+1} from F_k
    - **Candidate Pruning**: Prune candidate itemsets in L_{k+1} containing subsets of length k that are infrequent
    - **Support Counting**: Count the support of each candidate in L_{k+1} by scanning the DB
    - **Candidate Elimination**: Eliminate candidates in L_{k+1} that are infrequent, leaving only those that are frequent => F_{k+1}

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Factors Affecting Complexity of Apriori

- Choice of minimum support threshold
  - lowering support threshold results in more frequent itemsets
  - this may increase number of candidates and max length of frequent itemsets
- Dimensionality (number of items) of the data set
  -
- Size of database
  -
- Average transaction width
  -

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example

TID	Items
1	Bread, Milk
2	Beer, Bread, Diaper, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Bread, Coke, Diaper, Milk



Items (1-itemsets)

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Minimum Support = 3

If every subset is considered,

$${}^6C_1 + {}^6C_2 + {}^6C_3$$

$$6 + 15 + 20 = 41$$

With support-based pruning,

$$6 + 6 + 1 = 13$$

Minimum Support = 2

If every subset is considered,

$${}^6C_1 + {}^6C_2 + {}^6C_3 + {}^6C_4$$

$$6 + 15 + 20 + 15 = 56$$

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Factors Affecting Complexity of Apriori

- Choice of minimum support threshold
  - lowering support threshold results in more frequent itemsets
  - this may increase number of candidates and max length of frequent itemsets
- Dimensionality (number of items) of the data set
  - More space is needed to store support count of itemsets
  - if number of frequent itemsets also increases, both computation and I/O costs may also increase
- Size of database
- Average transaction width
  -

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Factors Affecting Complexity of Apriori

- Choice of minimum support threshold
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- Size of database
  - run time of algorithm increases with number of transactions
- Average transaction width

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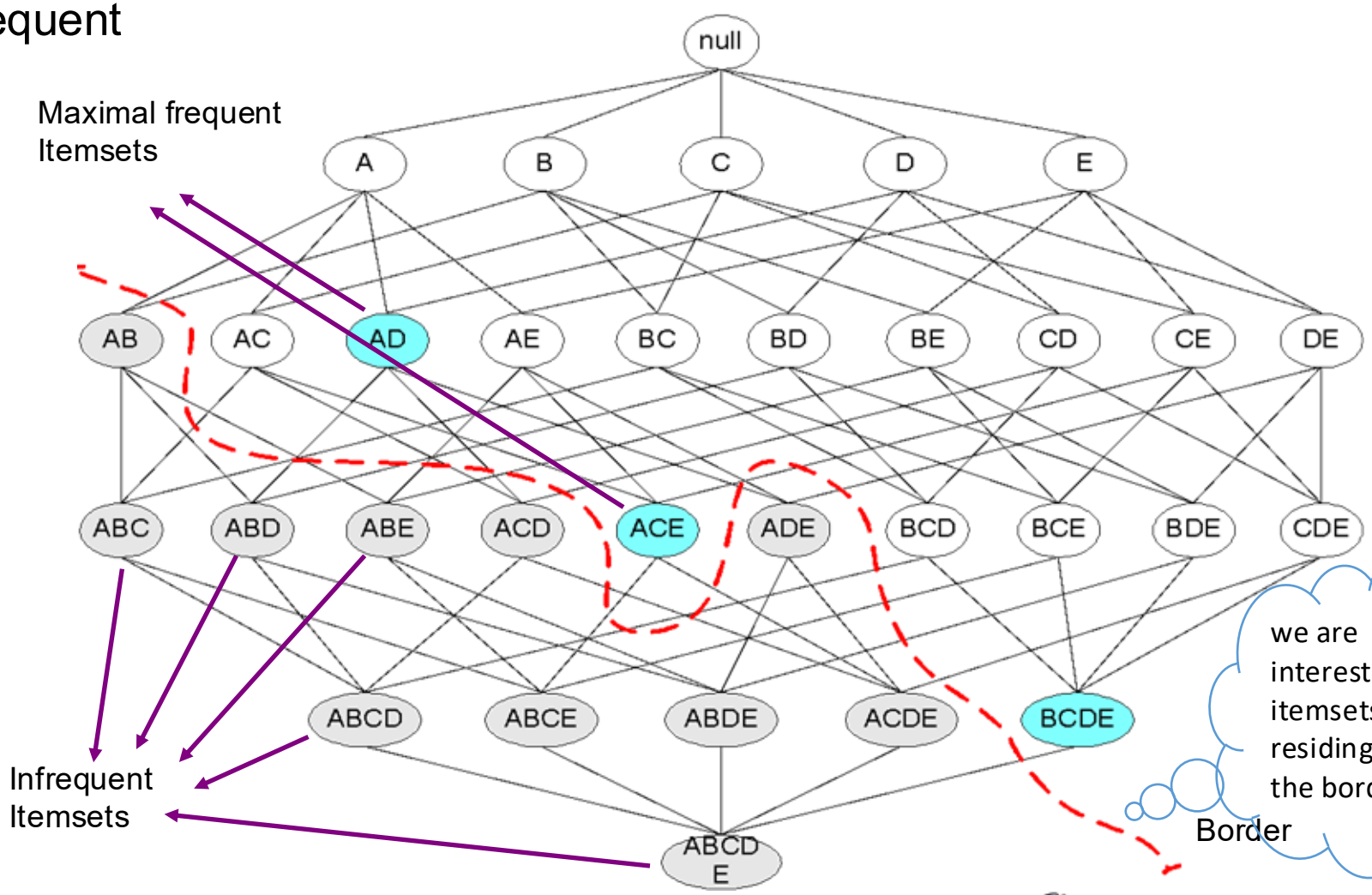
Factors Affecting Complexity of Apriori

- Choice of minimum support threshold
  - lowering support threshold results in more frequent itemsets
  - this may increase number of candidates and max length of frequent itemsets
- Dimensionality (number of items) of the data set
  - More space is needed to store support count of itemsets
  - if number of frequent itemsets also increases, both computation and I/O costs may also increase
- Size of database
  - run time of algorithm increases with number of transactions
- Average transaction width
  - transaction width increases the max length of frequent itemsets
  - This may increase max length of frequent itemsets and traversals of hash tree (number of subsets in a transaction increases with its width)

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Maximal Frequent Itemsets

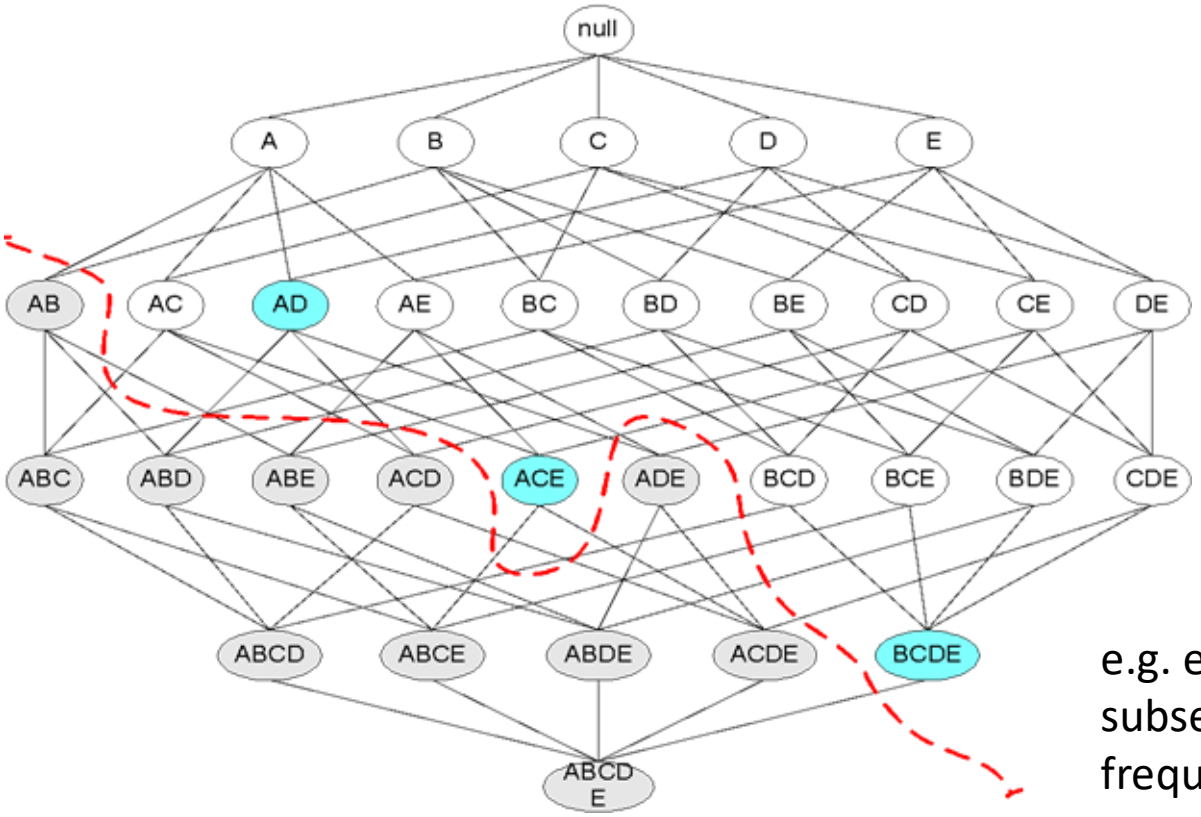
A frequent itemset is maximal if none of its immediate supersets are frequent



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Maximal Frequent Itemsets

A frequent itemset is maximal if none of its immediate supersets are frequent



The maximal frequent itemsets effectively provide a compact representation of frequent itemsets.
They form the smallest set of itemsets from which all frequent itemsets can be derived

e.g. every frequent itemset is a subset of one of the three maximal frequent itemsets, ad, ace, and bcde

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mimimal representation of all itemsets without losing their support information.

Closed Itemset

- An itemset X is closed if none of its immediate supersets has the same support as the itemset X.
- X is not closed if at least one of its immediate supersets has support count as X.

TID	Items
1	{A,B}
2	{B,C,D}
3	{A,B,C,D}
4	{A,B,D}
5	{A,B,C,D}

Itemset	Support
{A}	4
{B}	5
{C}	3
{D}	4
{A,B}	4
{A,C}	2
{A,D}	3
{B,C}	3
{B,D}	4
{C,D}	3

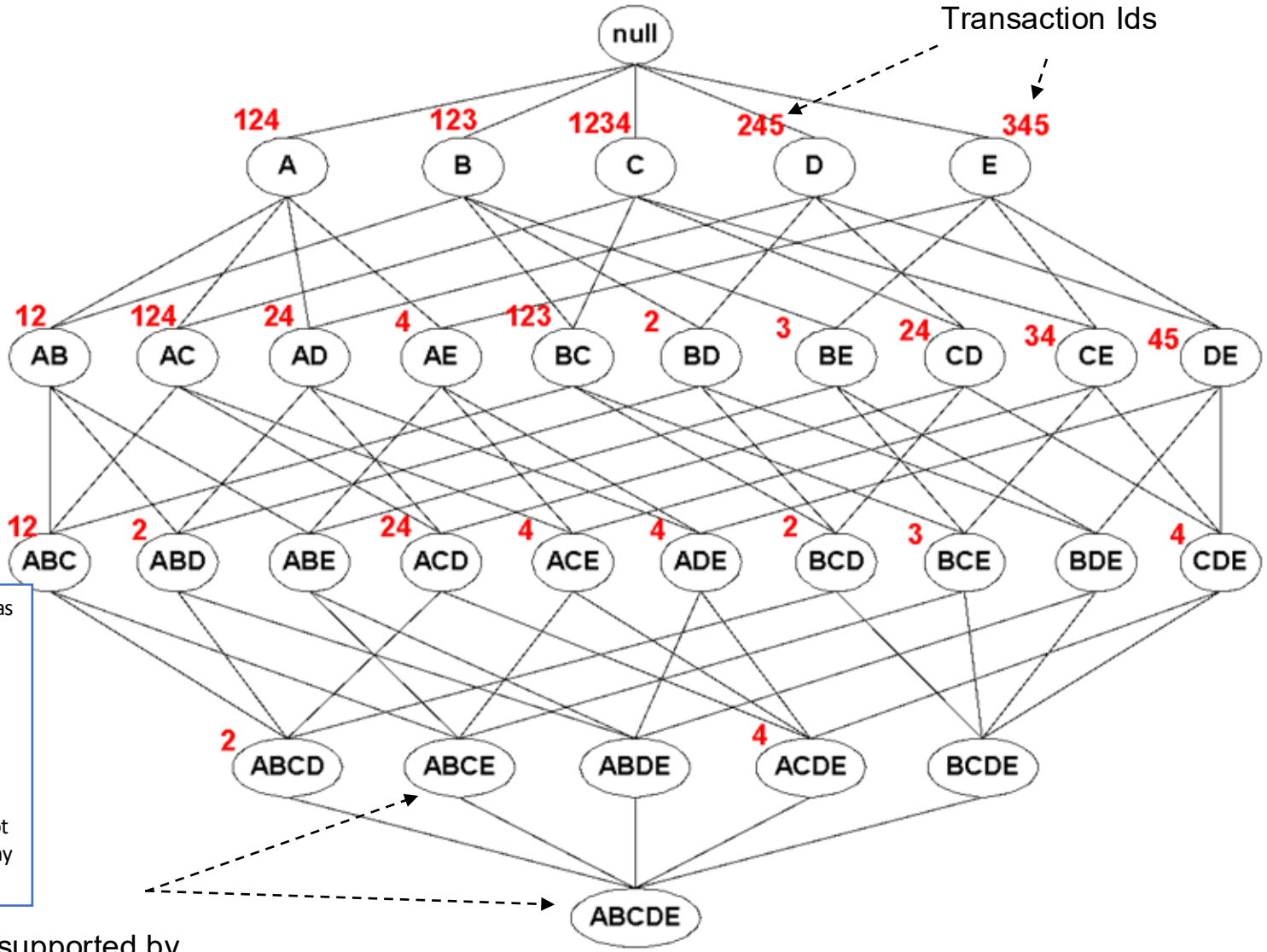
Itemset	Support
{A,B,C}	2
{A,B,D}	3
{A,C,D}	2
{B,C,D}	2
{A,B,C,D}	2

Example
Is {A} a closed itemset?
No. That's because the support of {A} and {A,B} are the same.

Example
Only {B} a closed itemset

Maximal vs Closed Itemsets

TID	Items
1	ABC
2	ABCD
3	BCE
4	ACDE
5	DE



B? AD? BC?

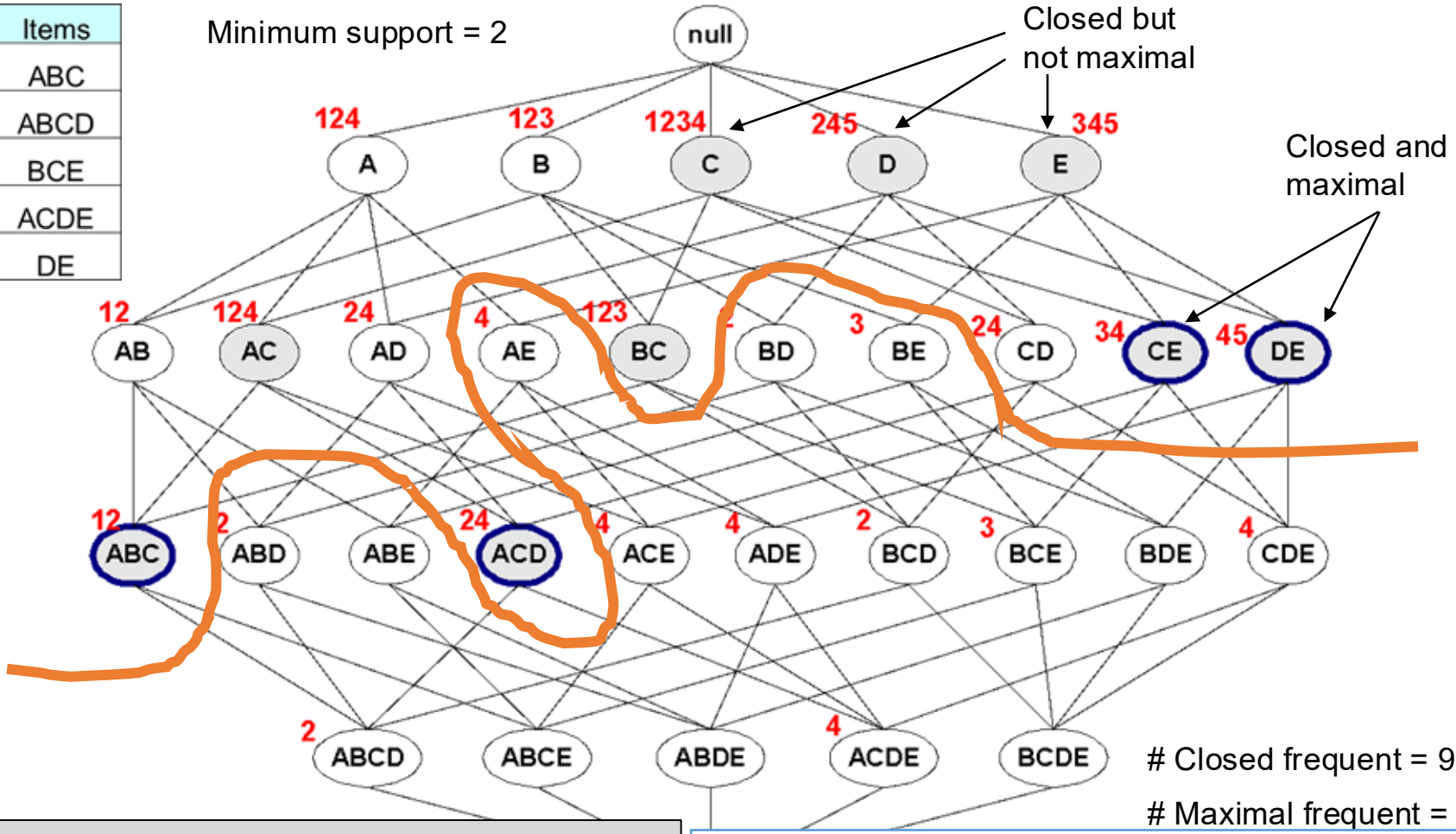
e.g. The support for b is the same as the support for bc every transactions that contains b also contains c. Thus, b is not a closed itemset.
e.g. Itemset ad is not closed because it has the same support count as its superset acd.
e.g. Bc is closed because it does not have the same support count as any of its supersets.

Not supported by any transactions

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Maximal Frequent vs Closed Frequent Itemsets

TID	Items
1	ABC
2	ABCD
3	BCE
4	ACDE
5	DE

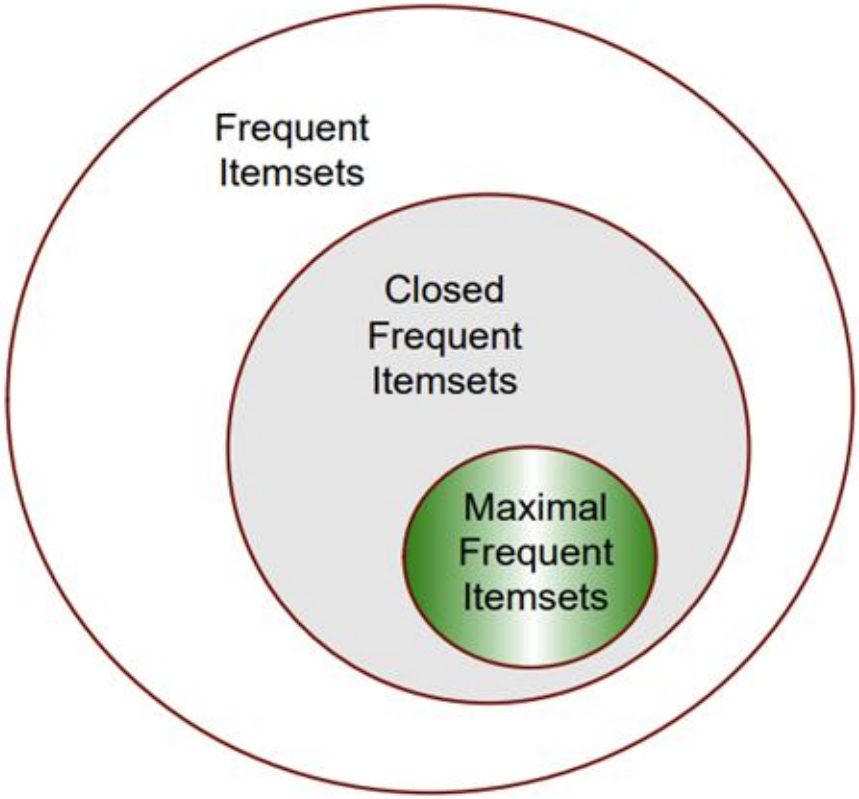


An itemset X is closed if none of its immediate supersets has the same support as the itemset X

Recall: an itemset is maximal frequent if none of its immediate supersets is frequent

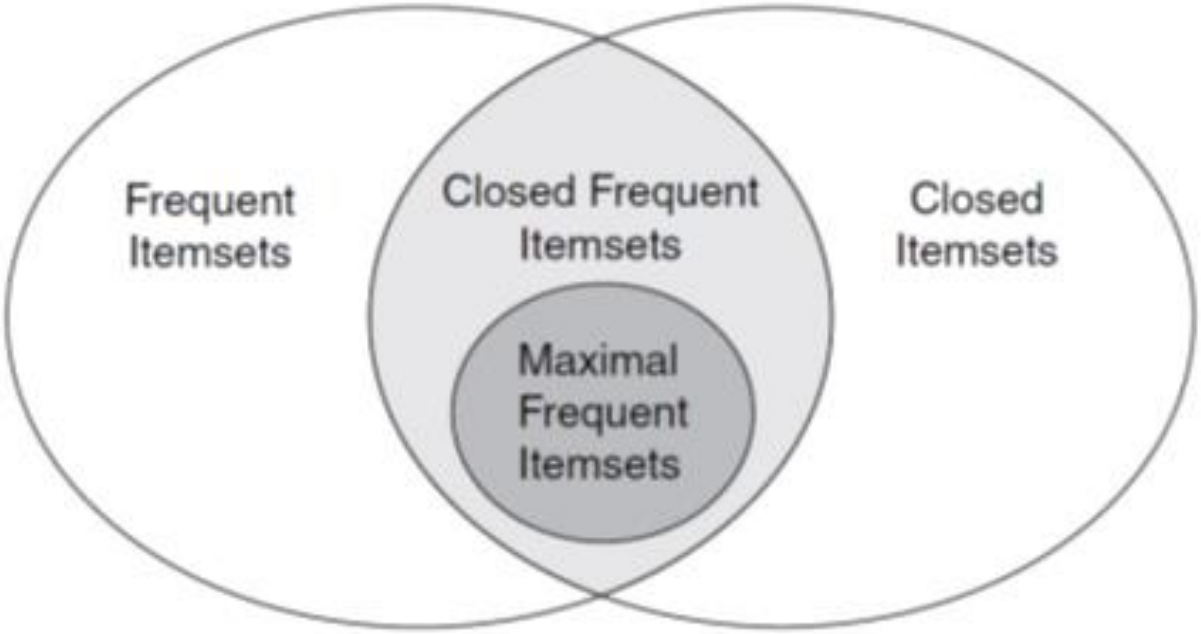
An interesting finding: all maximal frequent itemsets are closed because none of the maximal frequent itemsets can have the same support count as their immediate supersets.

Maximal vs Closed Itemsets



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Relationships among Closed, frequent, closed frequent, and maximal frequent itemset?



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SUMMARY

- Association rule mining basics
- Frequent itemset generation
- Compact representation
  - Maximal frequent itemset
  - Closed itemset

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Thank you

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